Please write clearly in block capitals. Centre number | Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE

Paper 2 (Generated practice paper - Set 13)

Assessed: The living environment, Biological resources, Sustainability, Research methods

Time allowed: 3 hours

Materials

For this paper you may use:
- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
  - use good English
  - organise information clearly
  - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
2	
3	
4	
5	
6	
7	
8	
9	
10	
11	
TOTAL	

Do not write outside the box

Answer all questions in the spaces provided.

0 1 . 1 Explain why food surpluses produced in MEDCs in the 1970s could not easily be sold.

[3 marks]

.....

.....

.....

.....

.....

.....

.....

0 1 . 2 Describe **four** methods used to reduce over-production while maintaining farm incomes.

[4 marks]

1 .....

.....

2 .....

.....

3 .....

.....

4 .....

.....

0 1 . 3 Describe **three** features that farmers may be paid for under an agri-environmental scheme and the conservation purpose of each.

[3 marks]

Do not write outside the box

1 .....

.....

2 .....

.....

3 .....

.....

10

0 2 . 1 Which of the following is a feature of natural systems? Tick **one** box.

[1 mark]

A Reactions that require high temperatures ☐

B Reliance on non-renewable energy ☐

C Waste products that are the raw materials for other processes ☐

D Dispersal of elements so they cannot be re-used ☐

0 2 . 2 Explain why nitrogen fixation by bacteria is more sustainable than the Haber process.

[2 marks]

.....

.....

.....

.....

0 2 . 3 Explain why incineration of wastes is less sustainable than natural decomposition.

[2 marks]

.....

.....

.....

.....

5

Do not write outside the box

0 3 . 1 Describe how cloning could be used to conserve the northern white rhinoceros, and explain why success has been limited.

[3 marks]

0 3 . 2 Explain how micro-propagation and cloning are used to conserve rare plants such as the Atlantic black poplar.

[3 marks]

0 3 . 3 Describe the purpose and methods of the Millennium Seed Bank.

[4 marks]

Do not write outside the box

10

0 4 . 1 Give **two** wild species with potential as new food species and, for each, one characteristic that makes it useful.

[2 marks]

1 .....

.....

2 .....

.....

0 4 . 2 Explain why indigenous species may be better for domestication than introduced livestock or crops.

[2 marks]

.....

.....

.....

.....

0 4 . 3 Explain how species of the spinach family could help to solve a problem caused by irrigation.

[1 mark]

.....

.....

5

0 5 Table 1 compares two exploited species.

Table 1

Feature	Anchovy	Blue whale
Age at sexual maturity	1 year	10 years
Offspring per female per year	Thousands of eggs	1 calf every 2-3 years
Lifespan	3-4 years	80+ years
Parental care	None	Extensive

0 5 . 1 Use Table 1 to identify which species is r-selected and which is k-selected, and explain the difference between the strategies.

[3 marks]

.....

Do not write outside the box

Do not write outside the box

0 5 . 2 Explain why k-selected species are much more easily over-exploited than r-selected species.

[3 marks]

0 5 . 3 State the equation used to forecast a change in population size.

[1 mark]

0 5 . 4 Explain what is meant by **maximum sustainable yield** and how it relates to carrying capacity.

[3 marks]

10

0 6 Table 1 compares an indigenous mixed broadleaf woodland with a Sitka spruce plantation of the same area in Scotland.

Table 1

Feature	Mixed indigenous woodland	Sitka spruce plantation
Tree species	12	1
Age structure	All ages, dead wood present	All planted in the same year
Planting density	Natural, variable	Close, 2 m spacing
Timber yield / m^3 ha^{-1} yr^{-1}	4	16
Bird species	38	9
Ground flora species	60	4

0 6 . 1 Use Table 1 to explain why the plantation has a much higher timber yield but far lower biodiversity.

[6 marks]

0 6 . 2 Explain why commercial plantations are felled before the trees reach full size.

[2 marks]

Do not write outside the box

Do not write outside the box

.....

.....

0 6 . 3 Explain why there is little demand today for coppiced or pollarded wood.

[2 marks]

.....

.....

.....

.....

10

0 7 . 1 Explain why tropical rainforests have such high biodiversity.

[4 marks]

.....

.....

.....

.....

.....

.....

.....

.....

0 7 . 2 Explain why rainforest species may be particularly vulnerable to environmental change.

[2 marks]

.....

.....

.....

.....

0 7 . 3 Explain why rainforest plants and animals such as frogs are a rich source of medicines.

[2 marks]

.....

Do not write outside the box

.....

.....

.....

0 7 . 4 Explain the importance of tropical rainforest for the hydrological cycle and for soils.

[2 marks]

.....

.....

.....

.....

10

0 8 . 1 Which feature of the Earth prevents most atmospheric gases from escaping into space? Tick **one** box.

[1 mark]

A Its mass ☐

B Its magnetic field ☐

C Its distance from the Sun ☐

D Its speed of rotation ☐

0 8 . 2 Explain how the speed of rotation of the Earth and the tilt of its axis affect the conditions for life.

[2 marks]

.....

.....

.....

.....

0 8 . 3 Explain why the presence of liquid water on Earth depends on **both** the distance from the Sun **and** atmospheric pressure.

[2 marks]

.....

.....

.....

.....

5

0 9 Table 1 shows the energy inputs and food energy outputs of three farming systems, in GJ ha^{-1} yr^{-1}.

Table 1

System	Energy input / GJ ha^{-1} yr^{-1}	Food energy output / GJ ha^{-1} yr^{-1}
Subsistence maize, hand cultivated	0.6	9.0
Intensive wheat, UK	18.0	54.0
Intensive beef, feedlot	40.0	12.0

0 9 . 1 Calculate the energy ratio (output / input) for each system in Table 1.

[3 marks]

Answer

0 9 . 2 Explain what is meant by an **energy subsidy**, and use your answers to explain why the intensive systems have much lower energy ratios.

[3 marks]

.....

.....

.....

.....

.....

.....

.....

0 9 . 3 Explain **three** ways a farmer can reduce the energy losses in a food production chain.

[3 marks]

Do not write outside the box

Do not write outside the box

0 9 . 4 Explain **two** advantages of genetic modification over conventional selective breeding.

[2 marks]

0 9 . 5 Explain **four** concerns about the use of GM crops.

[4 marks]

15

1 0 Table 1 shows the catch made by a shrimp trawler during one fishing trip.

Table 1

Component of catch	Mass / tonnes
Shrimps (target species)	0.6
Crabs, sea urchins, molluscs and starfish	5.9
Immature fish of commercial species	1.9

1 0 . 1 Calculate the percentage of the total catch that was by-catch.

[2 marks]

Answer ☐ %

1 0 . 2 Explain why shrimp trawling produces such a high proportion of by-catch.

[1 mark]

.....

.....

1 0 . 3 Explain why k-selected species such as the common skate and the Greenland shark are especially vulnerable to overfishing.

[3 marks]

.....

.....

.....

.....

.....

.....

.....

1 0 . 4 Evaluate the methods that can be used to reduce the environmental impacts of fishing.

[9 marks]

.....

.....

Do not write outside the box

Do not write outside the box

Extra space

Do not write outside the box

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

15

Write an essay on one of the following topics.

1 1 . 1 Discuss how ecological monitoring and the development of new technologies have improved the planning and evaluation of conservation strategies.

[25 marks]

OR

1 1 . 2 Discuss the extent to which the principles of the circular economy could be used to develop sustainable lifestyles.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

Do not write outside the box

Do not write outside the box

Do not write outside the box

Do not write outside the box

Do not write outside the box

END OF QUESTIONS